

## Bachelor's Thesis at the TU Berlin chair Industrial Information Technology

Fachgebiet Industrielle  
Informationstechnik

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### Title: Generation of Car-Cyclist Accident Scenarios Using Foundation Models

As cycling continues to grow as a mode of urban mobility, the safety of cyclists in mixed traffic becomes increasingly important. Validating automated driving functions for car-cyclist conflicts is a safety critical task. Real-world testing alone would require millions of kilometers to statistically demonstrate that an automated function performs at least as safely as a human driver in rare but critical situations.

Scenario-based testing addresses this limitation by systematically evaluating automated systems in simulation using curated traffic scenarios. The effectiveness of this approach depends directly on the quality, diversity and realism of the scenario catalogue, which is typically created manually using standardized formats such as OpenSCENARIO and OpenDRIVE. Recent advances in foundation models- large language models (LLM), vision-language models (VLMs) and diffusion-based generative models- provide new opportunities to automate and scale scenario generation by transforming real-world crash data and infrastructure descriptions into structured simulation inputs.

The objective of the thesis is to design, implement and evaluate a pipeline that uses foundation models to generate car-cyclist accident scenarios in standardized ASAM OpenSCENARIO and OpenDRIVE format. The generated scenarios will be parametrized using real-world crash datasets and demonstrated in a simulation environment. The thesis aims to provide both a structured comparison of foundation model approaches and crash datasets, as well as a working software prototype for scenario generation.

The scope of the bachelor's thesis includes the following tasks:

#### 1. Literature Review

- Analyze the state of the art in scenario generation using foundation models
- Compare different model types (e.g., LLMs, VLMs, generative models) and relevant adaptation techniques

#### 2. Dataset Analysis



- Identify and compare available car-cyclist accident datasets
- Analyze the contained parameters and assess their suitability for scenario generation
- Classify the relevant accident types between cars and cyclists

### **3. Scenario Parameter Mapping**

- Identify required parameters from OpenSCENARIO/OpenDRIVE
- Map dataset features to scenario parameters
- Define a structured representation for scenario generation

### **4. Model Selection and Pipeline Design**

- Select a foundation model and adaptation technique based on the literature review
- Design a pipeline architecture for generating scenarios from crash data
- Implement automatic generation of OpenSCENARIO and OpenDRIVE files

### **5. Simulation and Demonstration**

- Validate generated scenarios using esmini
- Demonstrate as least one generated scenario in Dyna4, including the scenario files, replacement of the ego-vehicle dynamics with the Dyna4 vehicle model and execution of the scenario

### **6. Evaluation**

- Define evaluation metrics covering syntactic validity, scenario diversity and criticality
- Apply the evaluation methodology to the generated scenarios

### **7. Documentation and Discussion**

- Document of the methodology, implementation and results
- Discuss limitations and challenges of using foundation models for scenario-based testing

## **Organisation**

The work is to be carried out in constant contact with the supervising assistant, Ms. Esra Demircioglu

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Prof. Dr.-Ing. Rainer Stark

The receipt of the problem statement and agreement to provide two copies of the thesis to the department are hereby confirmed.

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